



Hotel Operations Analytics

Marriott Case Study

Business Challenge:

Improve room readiness during peak hotel check-in.

Key Findings:

- Housekeeping capacity could not meet peak demand
- Staffing adjustments increased projected room readiness
- Operational changes improved service efficiency

Outcome:

Projected room readiness improvement from 68% to 79-80%

Business Challenge



Problem:

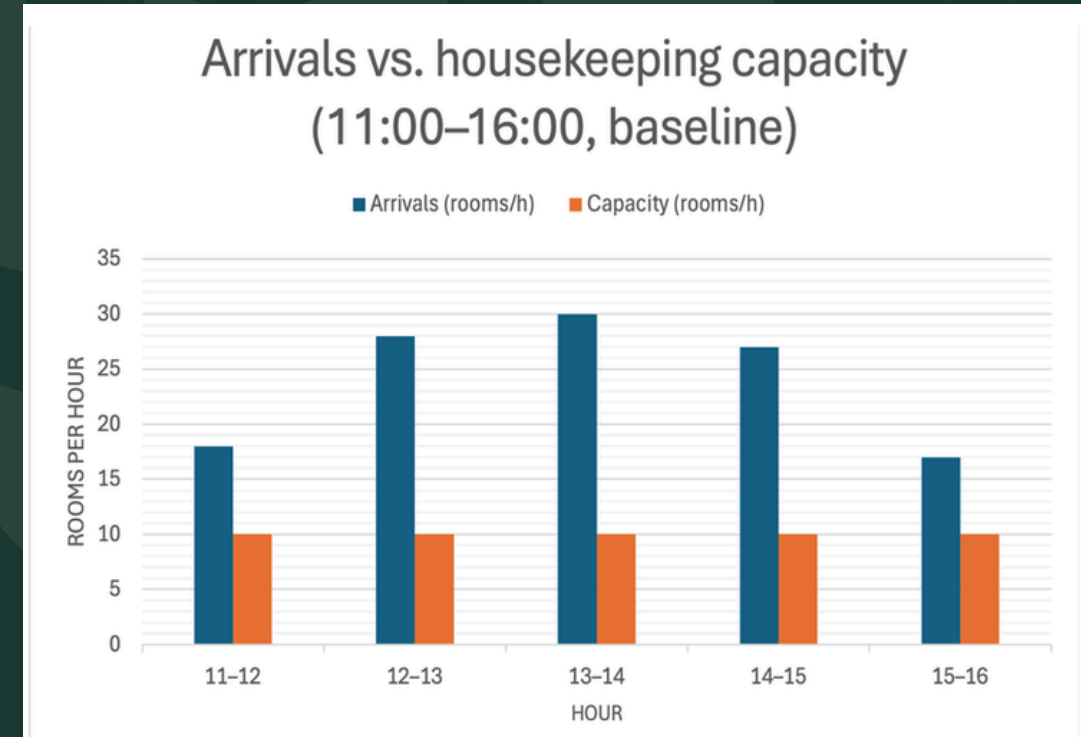
Marriott properties struggle to meet the 4:00 p.m. room readiness target because housekeeping capacity cannot keep pace with peak guest arrivals between 12:00–2:00 p.m.

Objective:

Improve operational efficiency and increase the percentage of rooms ready by 4:00 p.m.

My Approach:

- Built Excel-based capacity and operational models.
- Analyzed housekeeping capacity against guest arrival demand.
- Evaluated staffing, workflow, and arrival management scenarios.
- Compared operational alternatives to identify the most effective solution.
- Estimated the projected impact of each recommendation on room readiness.



Key Findings



Peak Demand

Housekeeping capacity could not meet demand during peak arrival hours.

Staffing

Increasing staffing during peak hours significantly improved projected room readiness

Operations

Small workflow improvements reduced bottlenecks without additional capital investment.

Recommendations

- Increase staffing during peak hours
- Standardize housekeeping workflows
- Smooth guest arrival demand
- Improve capacity planning

Business Impact

68%



79–80%

Projected room readiness improvement

Skills Used

- Excel
- Business Analytics
- Operations Analytics
- Capacity Planning
- Process Improvement
- Scenario Modeling
- Data Visualization
- Operational Strategy